

A Ray Tracing and FEM-based Simulation of Optical Waveguide-based Tactile Sensors

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Abstract—In this course project, ray tracing techniques were employed to simulate the propagation of light within the core of an optical waveguide, and the linear finite element method was utilized to model the deformation of the silicone rubber cladding. This was done to determine the optical power received by the photoelectric detectors (PDs). A tunable tactile sensor model based on silicone rubber and optical waveguides was proposed, which measures the force applied on the tactile sensor by the deformation of the silicone rubber, thereby causing a change in the optical waveguide, and thus a change in the light power received by the PD end. However, there is still room for improvement in the simulation accuracy, and the goal is to make the simulation results as close to the experimental data as possible.

Index Terms—Optical Waveguide, Tactile Sensor, Ray Tracing, Finite Element Simulation, Silicone Rubber, Optical Power

I. INTRODUCTION

In our previous research work in the laboratory, we proposed a flexible tactile sensor composed of silicone rubber and optical waveguides [1]. Its working principle is to embed optical waveguides in silicone rubber, with light emitting diodes (LEDs) and photoelectric detectors (PDs, specifically, photo triodes) embedded at the two ends of each optical waveguide core, providing light sources and detecting light powers, respectively (Fig. 1). When the silicone rubber undergoes deformation, the deformation of the embedded optical waveguide causes a change in light power received by the PD end, thereby measuring the force applied on the tactile sensor. The input of the tactile force sensor is force, and the output is the measured light power of PD.

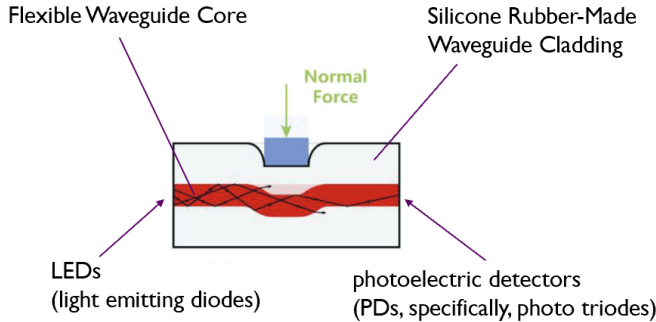


Fig. 1. Working principle of optical waveguide-based tactile sensors.

In this course project, we employed ray tracing techniques to simulate the propagation of light within the core of an optical waveguide. Additionally, we utilized the linear finite element method to model the deformation of the silicone rubber cladding. Ultimately, we were able to determine the optical power received by the photoelectric detectors (PDs).

II. TECHNICAL IMPLEMENTATION

A. Spatial discretization

In order to conveniently modify the shape of the sensor and the optical waveguide, we use pygmsh to generate a triangular mesh for the silicone rubber cladding. We generate the waveguide path using a third-order Bézier curve and discretize it into line segments (Fig. 2).

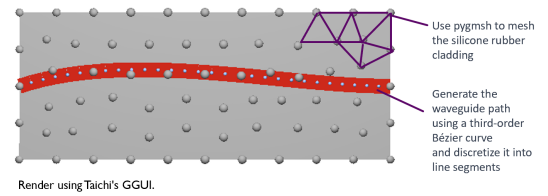


Fig. 2. Spatial discretization of tactile sensors.

B. Optical waveguide simulation

For the simulation of light propagation in the optical waveguide, we use a ray tracing algorithm. We consider three situations of light in the waveguide: total internal reflection, refraction and the effect of path length on light power. According to the Fresnel equations, as shown in Equation 1, when $k < 0$, total internal reflection of light occurs. Where: η : Represents the ratio of refractive indices (i.e., refractive index1/refractive index2). N : This is the surface normal vector. I : This is the incoming light direction vector.

$$k = 1.0 - \eta^2 * (1.0 - (N \cdot I)^2) \quad (1)$$

$$\begin{cases} F(\theta) = R_0 + (1 - R_0)(1 - \cos \theta)^5 \\ R_0 = \left(\frac{n_{\text{core}} - n_{\text{clad}}}{n_{\text{core}} + n_{\text{clad}}} \right)^2 \end{cases} \quad (2)$$

$$I = I_0 e^{-\alpha x} \quad (3)$$

When total internal reflection does not occur, the light beam is considered as a dielectric that only undergoes specular reflection and refraction, as shown in Equation 2. Schlick's approximation is used to calculate the proportion of reflected light, where R_0 is the reflectance at normal incidence, $F(\theta)$ is the Fresnel reflectance, n_{core} is the refractive index of the fiber core, n_{clad} is the refractive index of the fiber cladding, and θ is the angle between the observation direction and the surface normal. The ray tracing ends when refraction occurs, and continues when reflection occurs [2].

The Beer-Lambert Law is used to account for the attenuation of light power due to the length of the light path (Equation 3) [3]. Here, I is the intensity of light after it passes through a medium, I_0 is the initial intensity of light before it enters the medium, α is the absorption coefficient, indicating how strongly the medium absorbs light, and x is the distance the light has traveled through the medium. However, in this project, we found that the effect brought by this light path length is not significant.

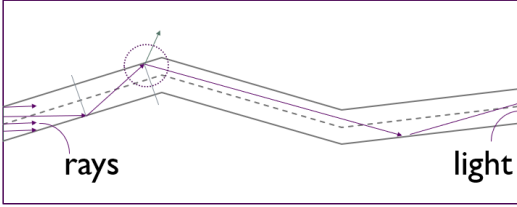


Fig. 3. Diagram of ray tracing.

The shape of the optical waveguide is discretized into two broken line segments. These line segments make it convenient to calculate the closest distance and normal direction. The rays are generated from the left end, their positions being random spots on the edge of the optical waveguide, but their direction is fixed, all moving horizontally to the right. The farthest right end is the light source, with the light power set at 100 (Fig. 3).

C. Linear Finite Element Simulation of Silicone Rubber Cladding

$$\Psi(F) = \frac{\mu}{2} \|\bar{F} - I\|_F^2 + \frac{\lambda}{2} \text{tr}(\bar{F} - I)^2 \quad (4)$$

For the simulation of the silicone rubber cladding, we use the corotated linear elasticity model with linear finite elements (Equation 4) [4]. Based on the calculated energy density, we employ the automatic differentiation system of Taichi to derive the forces. We update the velocities and displacements using the forward Euler method. we record the centroid coordinates of the initial discrete points of the optical waveguide. Finally, we perform interpolation for effective coupling with the optical waveguide.

III. RESULTS

As can be observed from the final simulation results, the light power changes as the tactile sensor deforms. The trend appears plausible, but there may still be a significant deviation

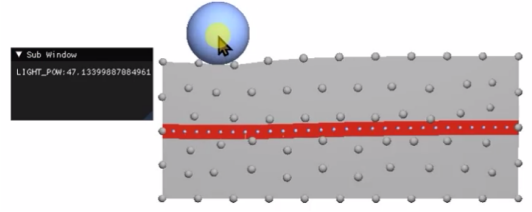


Fig. 4. Diagram of ray tracing.

from the actual experimental data. With 68 finite element nodes and 500 rays, the frame rate is approximately 7 FPS on a computer with a GTX1050 and i3-10100.

IV. DISCUSSIONS

Utilizing finite element analysis and ray tracing to simulate tactile sensors based on optical waveguides is proven to be feasible and shows potential for real-time application. However, there is still room for improvement in the accuracy of the simulation. Our aim is for it to eventually correspond closely with experimental data. Moving forward, our future plans involve expanding the simulator to encompass three dimensions and setting specific optimization objectives. This will be done in an attempt to optimize the design parameters of the tactile sensor, further advancing the potential applications of this simulation method.

ACKNOWLEDGMENTS

The code implementation for this course project refers to "Ray Tracing from Scratch" by HK-SHAO (<https://shao.fun/blog/w/taichi-ray-tracing.html>) and "Lecture 08 of Taichi Graphics Course S1: Elastic Object Simulation 01" by Tiantian Liu (<https://github.com/taichiCourse01/-Deformables>).

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